EECS 245, Fall 2026

LEC 17 Multiple Linear Regression; Gradients

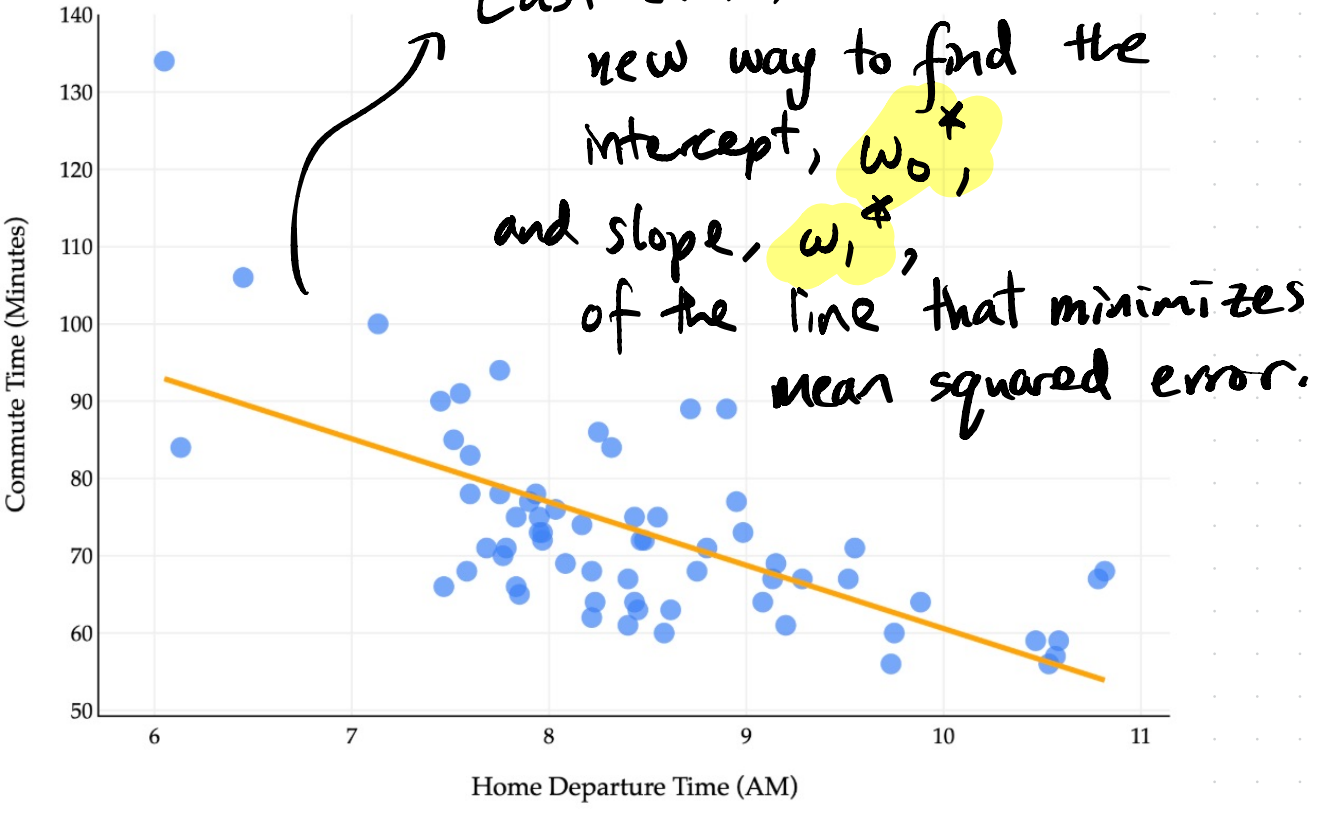
→ Read: Ch. 7.1, 7.2, 8.1

Agenda

1. Regression using linear algebra
   - → Recap: new formula for mean squared error
   - → Multiple linear regression
   - → Limitations
2. Gradients
   - → The extension of derivatives for functions $\mathbb{R}^d \to \mathbb{R}$

Announcements

- HW 7 due Sunday
  - → Includes survey: fill out after homework
- HW 6 solutions posted tomorrow
- Midterm 2 in 2 weeks from today!


Last class, we saw a new way to find the intercept, w_0^*, and slope, w_1^*, of the line that minimizes mean squared error.
Commute Time (Minutes)
140
130
120
110
100
90
80
70
60
50
6
7
8
9
10
11
Home Departure Time (AM)

new way of writing mean squared error

$$R_{sq}(\vec{w}) = \frac{1}{n} \| \vec{y} - X\vec{w} \|^2$$

error vector

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

"observation vector"

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}_{n \times 2}$$

"design matrix"

$$\vec{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

"parameter vector"

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

"observation vector"

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}_{n \times 2}$$

"design matrix"

$$\vec{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

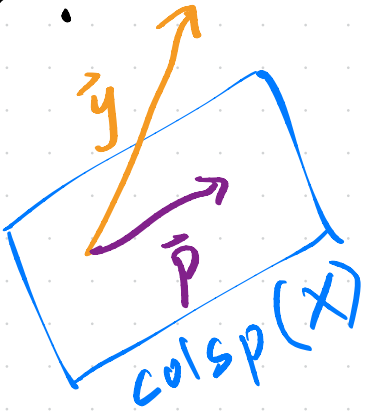
"parameter vector"

$$\vec{p} = X\vec{w} = \begin{bmatrix} w_0 + w_1 x_1 \\ w_0 + w_1 x_2 \\ \vdots \\ w_0 + w_1 x_n \end{bmatrix}$$

"prediction vector"

$$R_{sq}(\vec{w}) = \frac{1}{n} \| \vec{y} - X\vec{w} \|^2$$

Which $\vec{w}^*$ minimizes mean squared error?



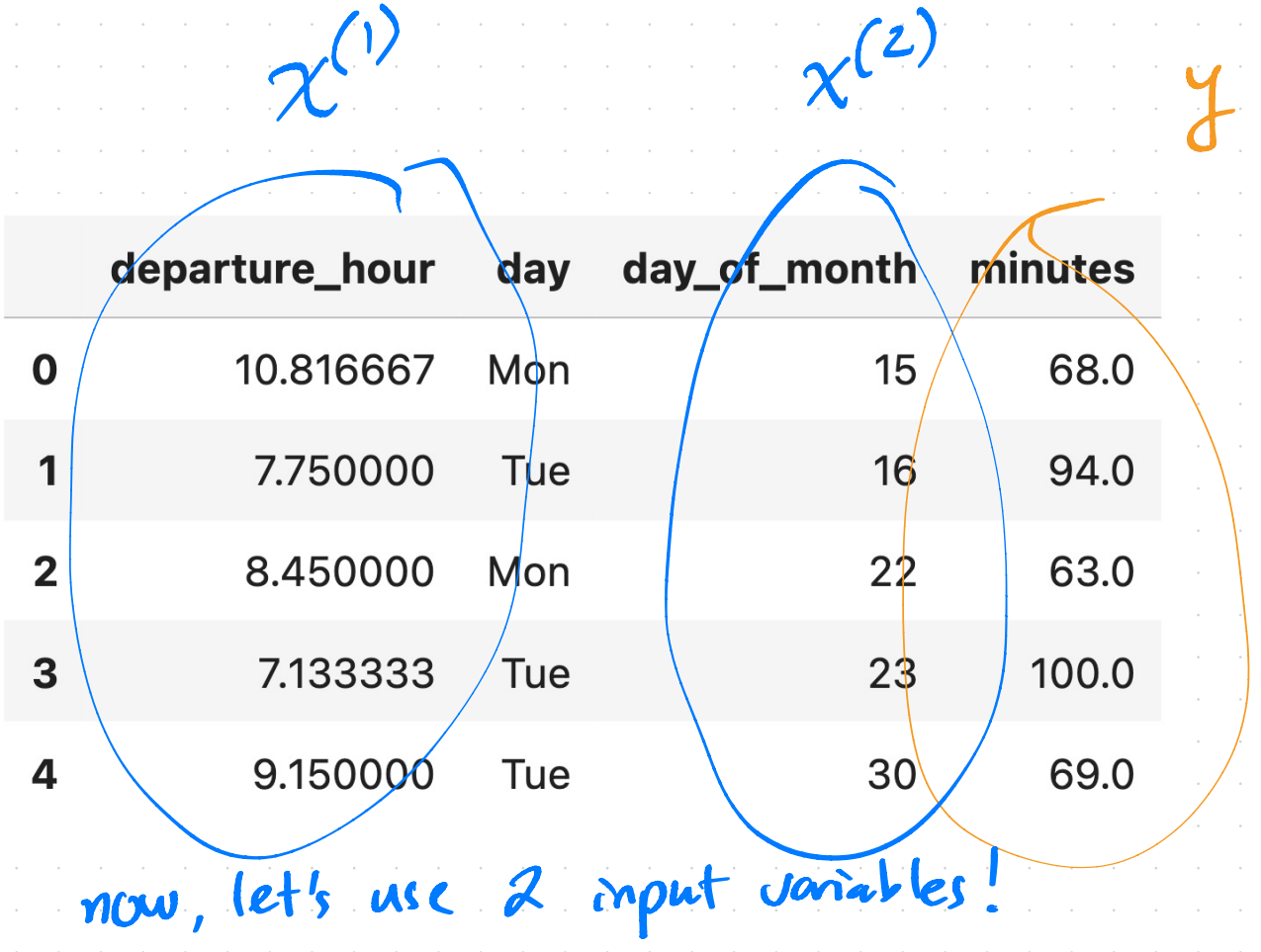
$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$$

requires $X^T X$ to be invertible, which requires X's cols to be linearly independent

⇒ otherwise, infinitely many solutions $\vec{w}^*$, all leading to same $\vec{p}$

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y} = \begin{bmatrix} \bar{y} - w_1^* \bar{x} \\ r \frac{\sigma_y}{\sigma_x} \end{bmatrix}$$

same formulas as before!!!



	departure_hour	day	day_of_month	minutes
0	10.816667	Mon	15	68.0
1	7.750000	Tue	16	94.0
2	8.450000	Mon	22	63.0
3	7.133333	Tue	23	100.0
4	9.150000	Tue	30	69.0

before

$$h(x_i) = w_0 + w_1 x_i$$

departure hour

now

$$h(\vec{x}_i) = w_0 + w_1 x_i^{(1)} + w_2 x_i^{(2)}$$

departure hour

day of month

multiple linear regression : multiple features

how do we find w_0^*, w_1^*, w_2^* ?

departure hour

day of month

$$h(\vec{x}_i) = w_0 + w_1 x_i^{(1)} + w_2 x_i^{(2)}$$

not squared?

$$\vec{w} = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

	departure_hour	day	day_of_month	minutes
0	10.816667	Mon	15	68.0
1	7.750000	Tue	16	94.0
2	8.450000	Mon	22	63.0
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4	9.150000	Tue	30	69.0

$$X_{n \times 3} = \begin{bmatrix} 1 & 10.816.. & 15 \\ 1 & 7.75 & 16 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

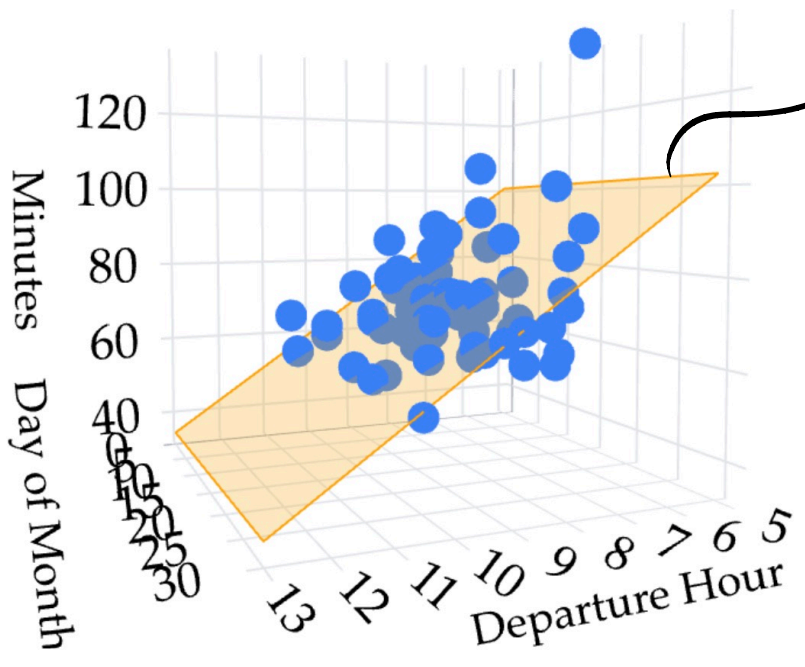
$$\vec{x}_1 = \begin{bmatrix} 10.816.. \\ 15 \end{bmatrix}$$

$$Aug(\vec{x}_1) = \begin{bmatrix} 1 \\ 10.816.. \\ 15 \end{bmatrix}$$

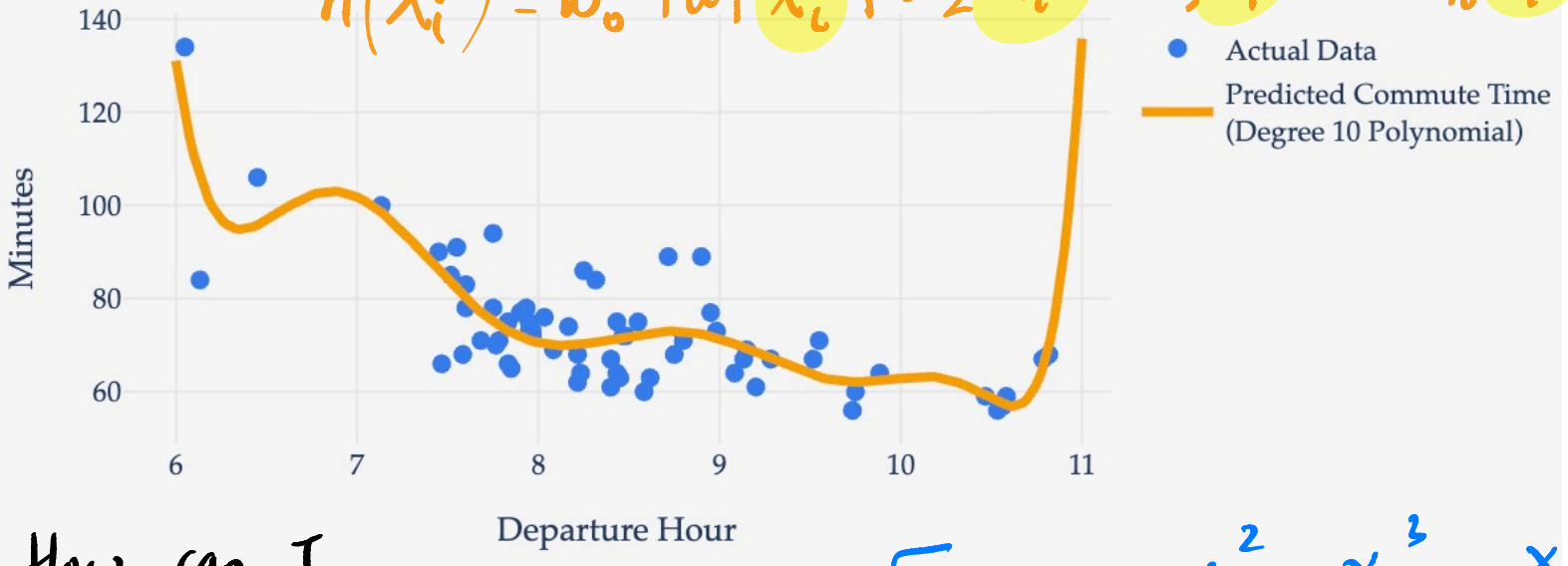
first 2 rows of design matrix

optimal parameters:

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y} \, !$$


120
100
80
60
40
30
Minutes
Day of Month
13
12
11
10
9
8
7
6
5
Departure Hour
affine subspace

$$h(x_i) = w_0 + w_1 x_i + w_2 x_i^2 + w_3 x_i^3 + \cdots + w_{10} x_i^{10}$$



How can I find the best fitting degree-10 poly using linear regression???

$$\underset{n \times 11}{X} = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 & \cdots & x_1^{10} \\ 1 & x_2 & x_2^2 & x_2^3 & \cdots & x_2^{10} \\ \vdots & \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & x_n^3 & \cdots & x_n^{10} \end{bmatrix}$$

Any model that is a linear combination of features can be fit using multiple linear regression, i.e.

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$$

$$h(x_i^{(1)}, x_i^{(2)}) = w_0 + w_1 \sin\left(x_i^{(1)} x_i^{(2)}\right) + w_2 \left(x_i^{(1)}\right)^{17} + w_3 e^{x_i^{(2)}}$$

→ good!

$$h(x_i^{(1)}, x_i^{(2)}) = w_0 + \sin\left(w_1 x_i^{(1)}\right)$$

w_1 within sin ⇒ bad!

Notation

In general: n data points, d features

$$h(\vec{x}_i) = w_0 + w_1 x_i^{(1)} + w_2 x_i^{(2)} + \dots + w_d x_i^{(d)}$$

$$= \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix} \cdot \begin{bmatrix} 1 \\ x_i^{(1)} \\ \vdots \\ x_i^{(d)} \end{bmatrix}$$

$\vec{w} \in \mathbb{R}^{d+1}$ parameter vector

$\vec{x}_i \in \mathbb{R}^d$ feature vector

$Aug(\vec{x}_i) \in \mathbb{R}^{d+1}$

$$= \vec{w} \cdot \underbrace{Aug(\vec{x}_i)}_{\text{rows of design matrix}}$$

$\times$ design matrix: n rows, $d+1$ cols!

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0	10.816667	Mon	15	68.0
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solution to using categorical data: one hot encoding

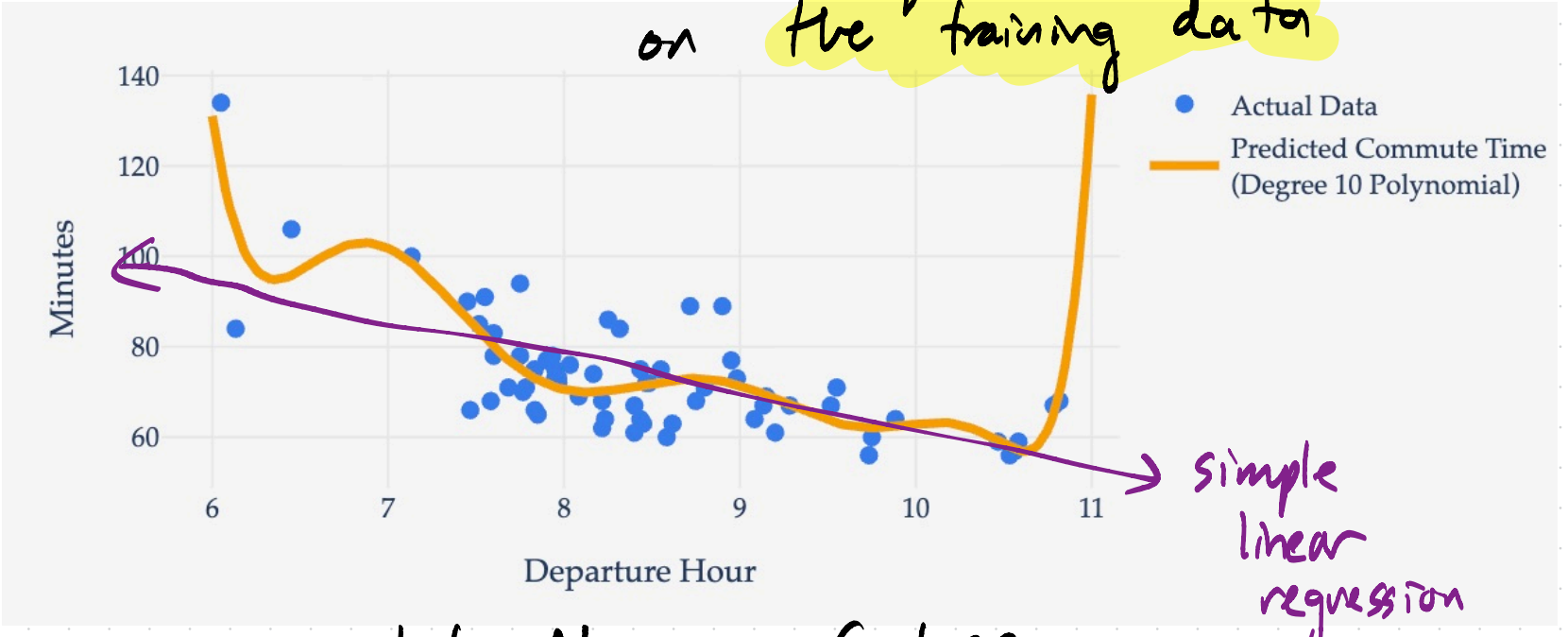
issue with linear independence

if intercept: 1 1 1 ... 1

day	day == Mon	day == Tue	day == Wed	day == Thu	day == Fri
Mon	1	0	0	0	0
Tue	0	1	0	0	0
Mon	1	0	0	0	0
...	...	...	...	...	...
Mon	1	0	0	0	0
Tue	0	1	0	0	0
Thu	0	0	0	1	0

=> sum 1 1 1 ... 1

adding more features can never increase mean squared error on the training data

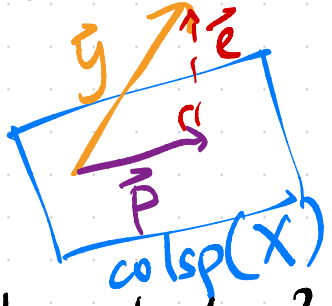


→ but adding more features can cause overfitting ⇒ worse performance on unseen data

To find $\vec{w}^*$ in multiple linear regression, we minimized mean squared error:

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

→ so far, we used a geometric argument

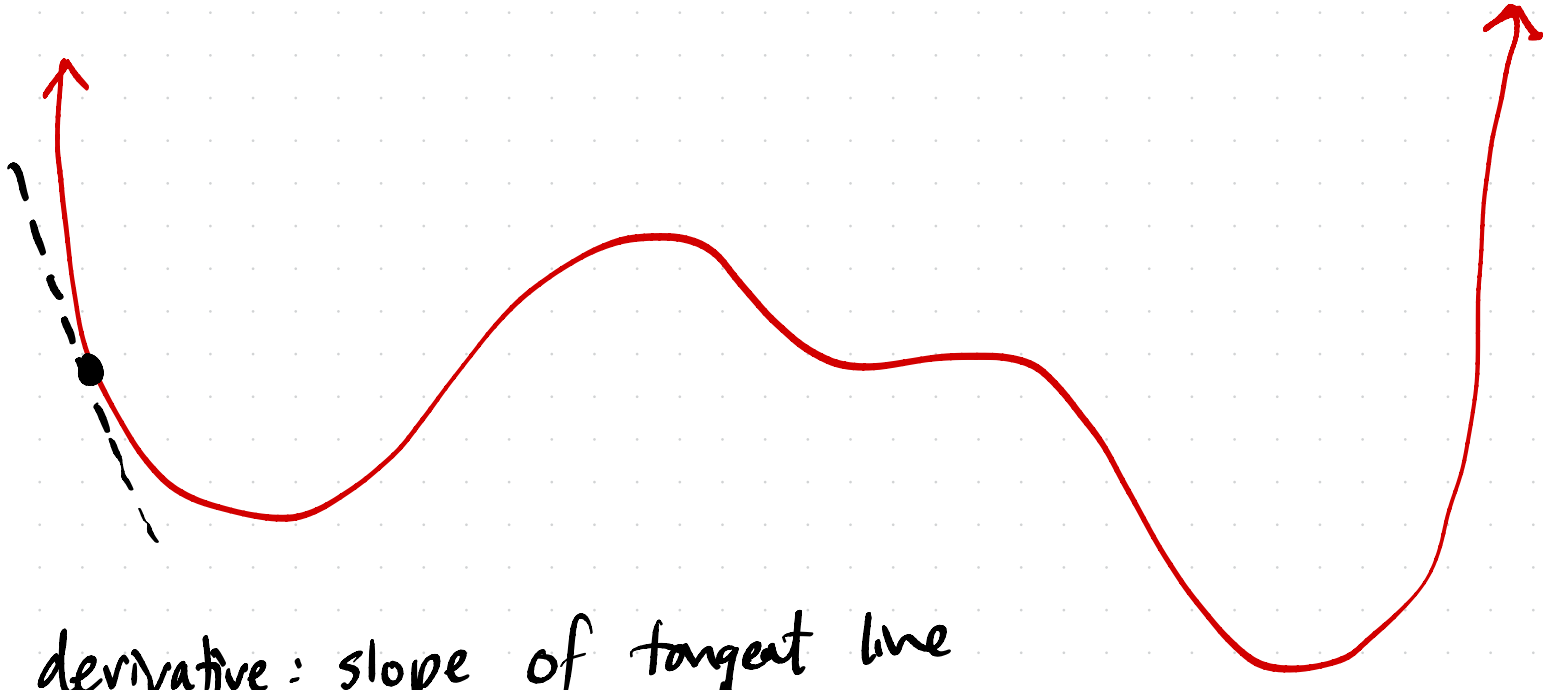


→ what about calculus?

$$R_{sq}(\vec{w}) = \frac{1}{n} \|\vec{y} - X\vec{w}\|^2$$

$$R_{sq} : \mathbb{R}^{d+1} \longrightarrow \mathbb{R}$$

"vector-to-scalar function"



derivative : slope of tangent line

$$f(\vec{x}) = x_1^2 + x_2^2 - 3x_1x_2 \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow \frac{\partial f}{\partial x_1} = 2x_1 - 3x_2$$

$$\Rightarrow \frac{\partial f}{\partial x_2} = 2x_2 - 3x_1$$

$$\nabla f(\vec{x}) = \begin{bmatrix} 2x_1 - 3x_2 \\ 2x_2 - 3x_1 \end{bmatrix}$$

e.g. $\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$\nabla f\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

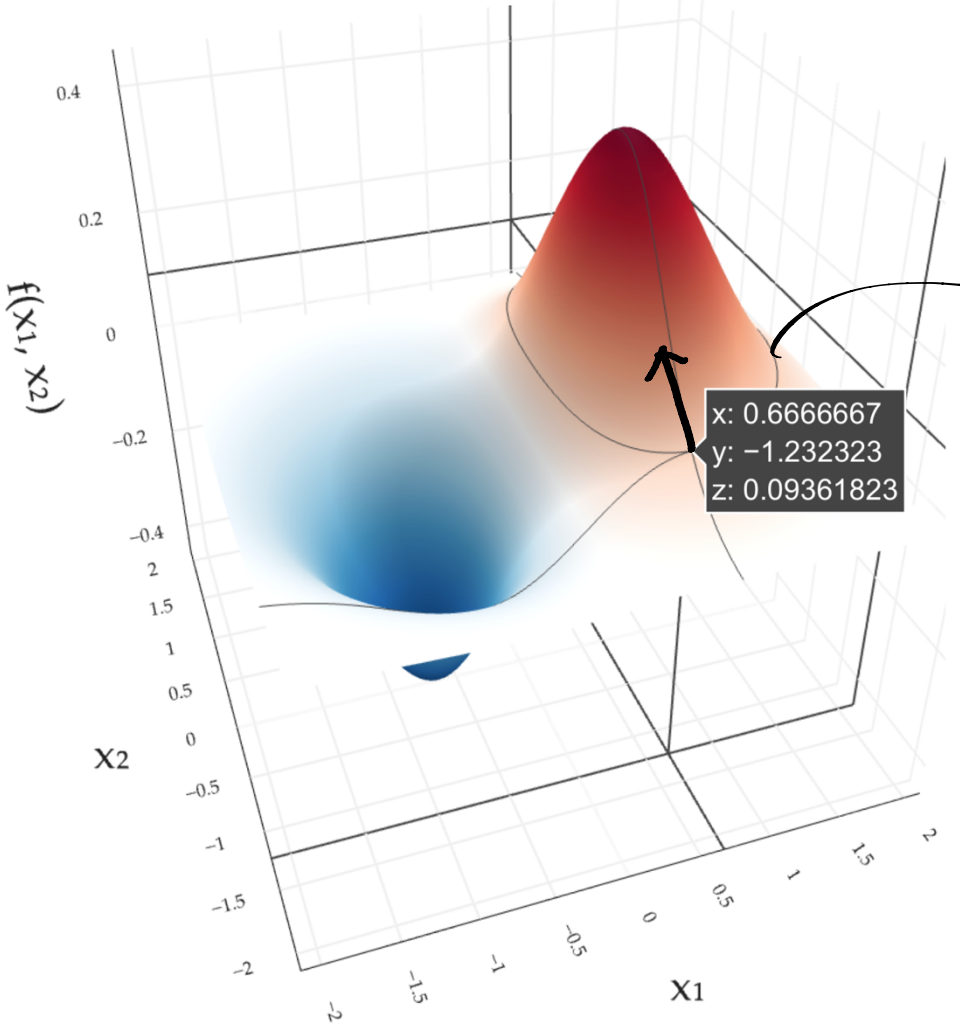
At the input $\vec{x}_0$,

the gradient

$$\nabla f(\vec{x}_0)$$

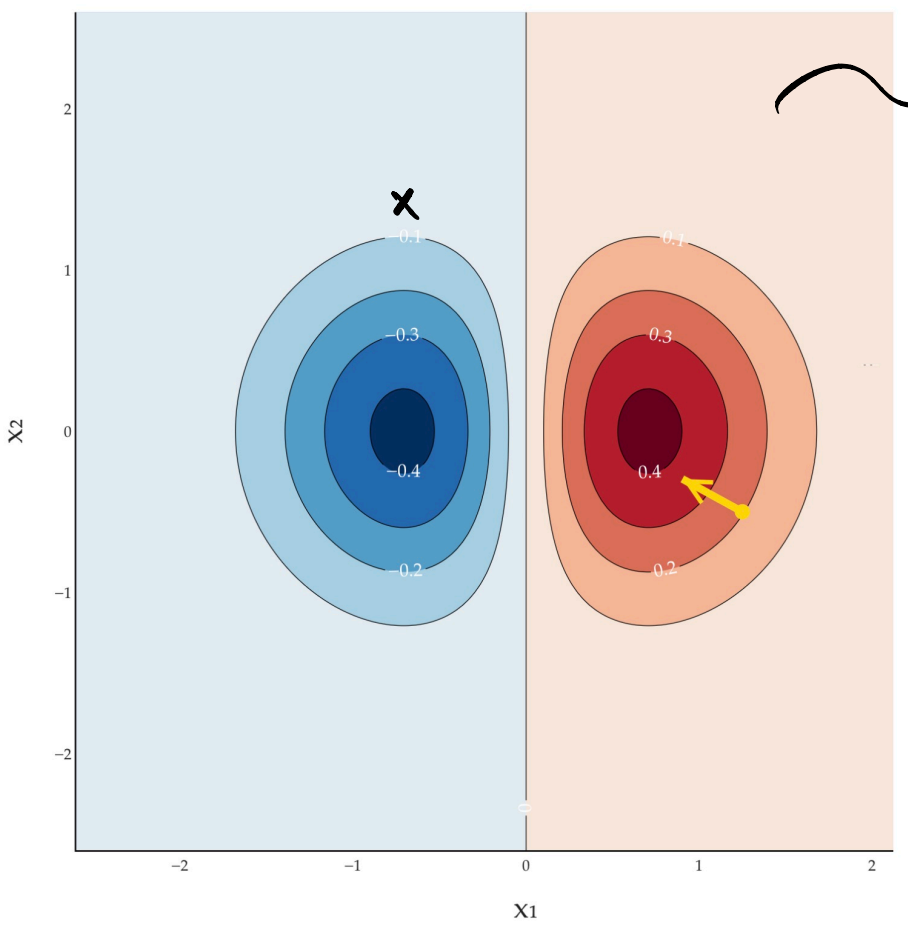
is the direction

of steepest increase!!!



gradient describes direction of steepest increase!

Gradient Vector at Point (1.25, -0.5)



contour plot

Chapter 8.2: "Big 3 Rules"

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{bmatrix}$$

① $$f(\vec{x}) = \vec{a} \cdot \vec{x}$$

$$= a_1 x_1 + a_2 x_2 + \cdots + a_d x_d$$

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_d \end{bmatrix} = \vec{a}$$

② $f(\vec{x}) = \|\vec{x}\|^2 = x_1^2 + x_2^2 + \cdots + x_d^2$

$$\frac{\partial f}{\partial x_i} = 2x_i$$

$$\nabla f(\vec{x}) = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_d \end{bmatrix} = 2\vec{x}$$